

Assignment 4

1. Text Cleaning

In this step, raw text data is cleaned by removing HTML tags, punctuation, and converting all text to lowercase. This ensures consistency and removes noise that does not contribute to semantic meaning.

2. Tokenization

Tokenization splits the cleaned text into individual words or tokens. This is important because NLP models process text as tokens rather than full sentences.

3. Stopword Removal

Common words like 'the', 'is', and 'and' are removed as they do not add significant meaning to the text and can reduce model efficiency.

4. Lemmatization

Words are reduced to their base or root form (e.g., 'running' → 'run'). This helps in treating different forms of a word as the same.

5. Vocabulary Creation

A vocabulary is created by collecting all unique words from the dataset. This helps in understanding word frequency and importance.

6. Bag of Words / TF-IDF

Text is converted into numerical format. Bag of Words counts word occurrences, while TF-IDF weighs words based on importance across documents.

7. Word Embeddings

Word2Vec is used to convert words into dense vectors that capture semantic relationships between words.

8. Sentence Embeddings

Sentence embeddings are created by averaging word vectors, allowing entire sentences to be represented numerically.

9. Similarity Search

Cosine similarity is used to compare sentence embeddings and find similar texts based on meaning.

10. Insights & Visualization

Visualizations like word frequency graphs and similarity heatmaps help in understanding data patterns and relationships.

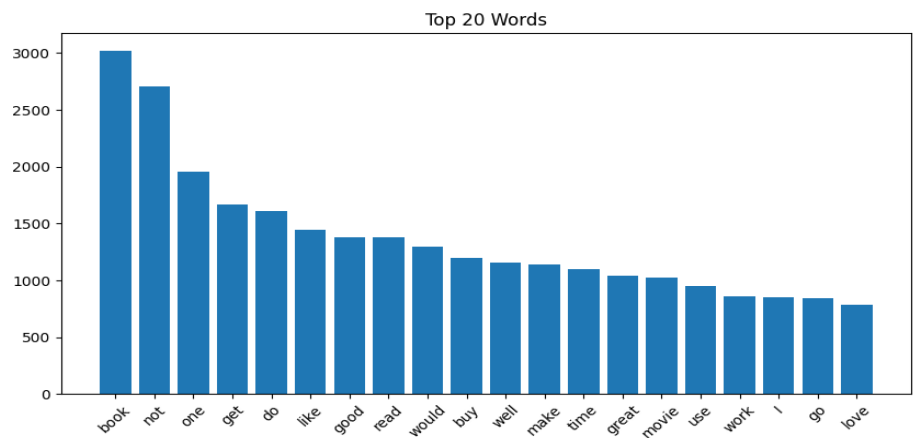


Figure 1: Output Visualization

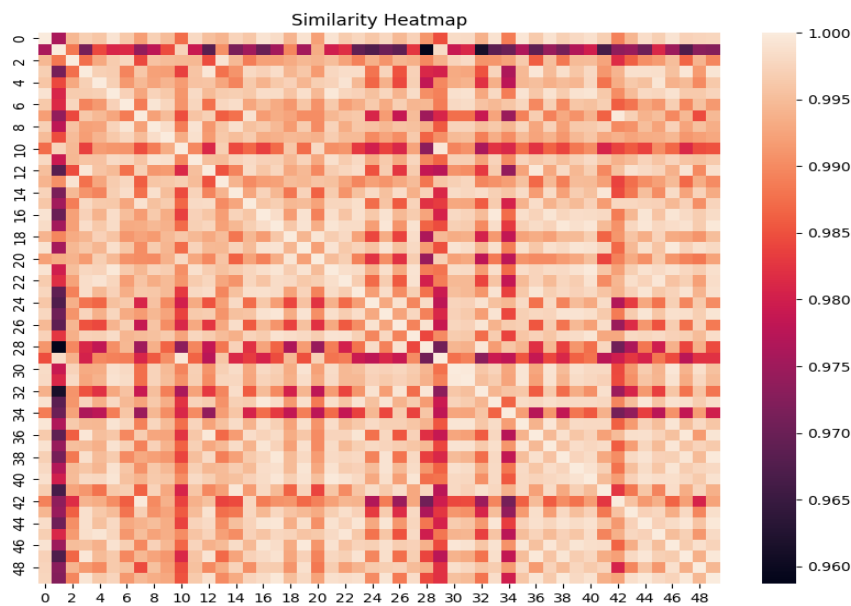


Figure 2: Output Visualization